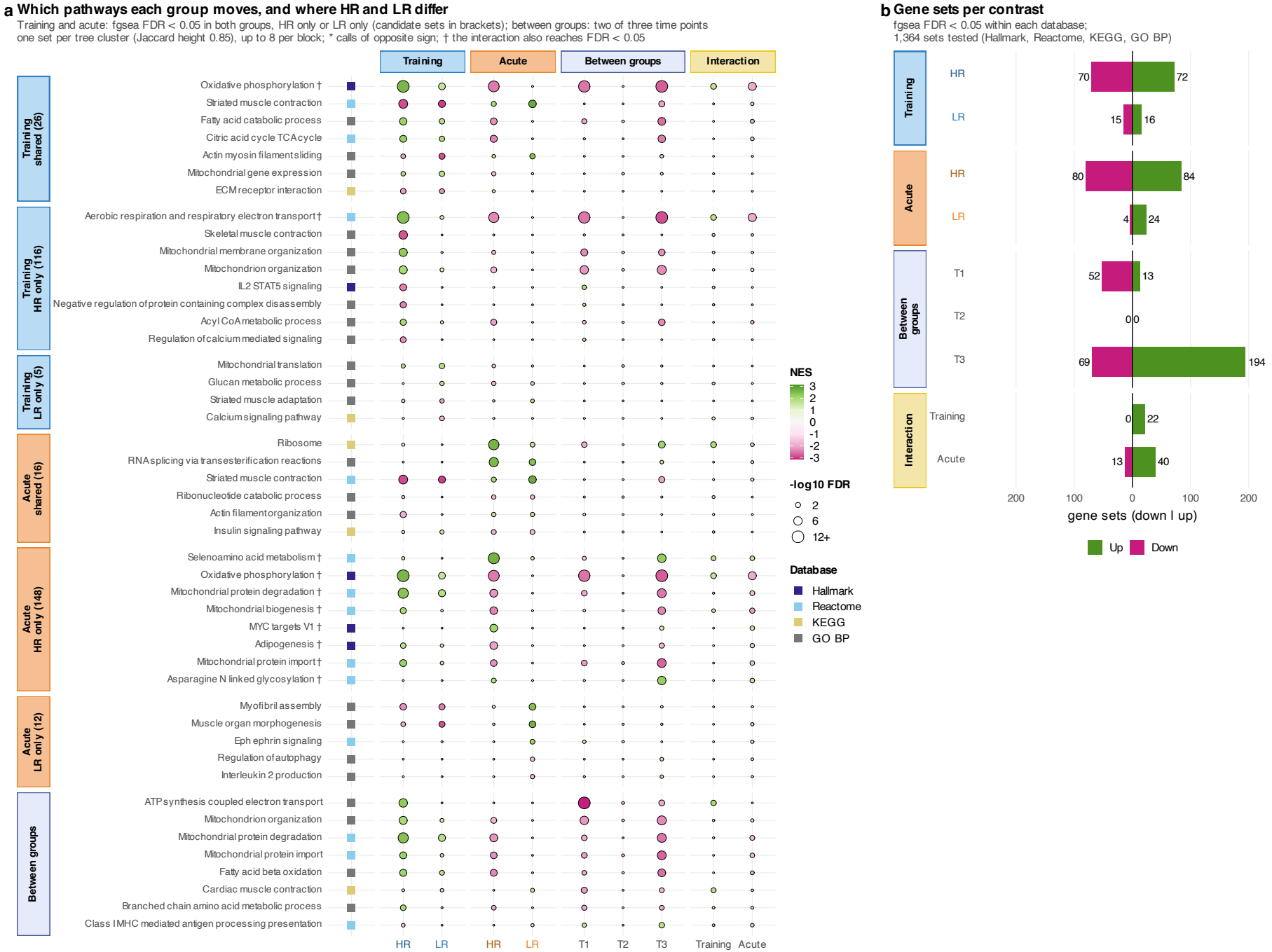
a Which pathways each group moves, and where HR and LR differ
Training and acute: fgsea FDR < 0.05 in both groups, HR only or LR only (candidate sets in brackets); between groups: two of three time points
one set per tree cluster (Jaccard height 0.85), up to 8 per block; * calls of opposite sign; † the interaction also reaches FDR < 0.05
Training
Acute
Between groups
Interaction
Training shared (26)
Oxidative phosphorylation †
Striated muscle contraction
Fatty acid catabolic process
Citric acid cycle TCA cycle
Actin myosin filaments sliding
Mitochondrial gene expression
ECM receptor interaction
Training HR only (116)
Aerobic respiration and respiratory electron transport †
Skeletal muscle contraction
Mitochondrial membrane organization
Mitochondrion organization
IL2 STAT5 signaling
Negative regulation of protein containing complex disassembly
Acyl CoA metabolic process
Regulation of calcium mediated signaling
Training LR only (5)
Mitochondrial translation
Glucan metabolic process
Striated muscle adaptation
Calcium signaling pathway
Acute shared (16)
Ribosome
RNA splicing via transesterification reactions
Striated muscle contraction
Ribonucleotide catabolic process
Actin filament organization
Insulin signaling pathway
Acute HR only (148)
Selenoamino acid metabolism †
Oxidative phosphorylation †
Mitochondrial protein degradation †
Mitochondrial biogenesis †
MYC targets V1 †
Adipogenesis †
Mitochondrial protein import †
Asparagine N linked glycosylation †
Acute LR only (12)
Myofibril assembly
Muscle organ morphogenesis
Eph ephrin signaling
Regulation of autophagy
Interleukin 2 production
Between groups
ATP synthesis coupled electron transport
Mitochondrion organization
Mitochondrial protein degradation
Mitochondrial protein import
Fatty acid beta oxidation
Cardiac muscle contraction
Branched chain amino acid metabolic process
Class IMHC mediated antigen processing presentation
HR
LR
HR
LR
T1
T2
T3
Training
Acute
NES
3
2
1
0
-1
-2
-3
-log10 FDR
2
6
12+
Database
Hallmark
Reactome
KEGG
GO BP
b Gene sets per contrast
fgsea FDR < 0.05 within each database;
1,364 sets tested (Hallmark, Reactome, KEGG, GO BP)
Training
HR
LR
70
72
15
16
Acute
HR
LR
80
84
4
24
Between groups
T1
T2
T3
52
13
0
0
69
194
Interaction
Training
Acute
0
22
13
40
200
100
0
100
200
gene sets (down l up)
Up
Down